



## Koneru Lakshmaiah Education Foundation

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### STUDENT CASE STUDY EXECUTION REPORT

|                |                          |
|----------------|--------------------------|
| Department     | FED                      |
| Subject        | DSA-2                    |
| Academic Year  | I-III Semester (2025-26) |
| Faculty Name   | Swetha Nagasri           |
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| Branch         | CSE                      |
| Course Outcome | CO2                      |

#### Case Study Topic: MongoDB Compound Index using B+ Tree on (user\_id, timestamp)

##### Work Done Summary:

In this case study, we analyzed how MongoDB uses a composite B+ Tree index on the fields (user\_id, timestamp). The objective was to determine whether different query types can efficiently utilize the compound index and to estimate the cost of a query involving an exact user\_id match and a timestamp range.

The study also examined why MongoDB may avoid using the compound index for queries that only specify the timestamp field without the prefix field user\_id.

##### Implementation Details:

- Considered a MongoDB collection indexed on (user\_id, timestamp).
- Analyzed three query patterns:
  - user\_id = 42 AND timestamp range
  - user\_id = 42
  - timestamp only
- Applied the B+ Tree prefix rule to determine index usability.
- Estimated B+ Tree height for approximately  $2 \times 10^9$  entries.
- Calculated the page reads required for the range query.

- Studied why suffix-only queries are inefficient.
- Concluded that prefix fields must be included for effective index utilization.

Result: Screenshots / Output

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MongoDB Compound Index Analysis

Query 1: user_id = 42 AND timestamp range
Index Usable: Yes

Query 2: user_id = 42
Index Usable: Yes

Query 3: timestamp only
Index Usable: No (Inefficient)

Estimated B+ Tree Height = 5
Approximate Cost = 4 page reads

Conclusion:
Prefix field user_id must be used for efficient index access.
```

Composite Index Usage

| Query Type                       | Index Usable?    | Reason                                           |
|----------------------------------|------------------|--------------------------------------------------|
| user_id = 42 AND timestamp range | Yes              | Uses prefix field user_id and range on timestamp |
| user_id = 42                     | Yes              | Index is sorted by user_id first                 |
| timestamp only                   | No (Inefficient) | Timestamp values are spread across all users     |

Cost Estimation

For:  
user\_id = 42  
timestamp BETWEEN '2024-01-01' AND '2024-01-31'  
Total entries:  $2 \times 10^9$   
B+ Tree Height:  $\log_{100}(2 \times 10^9) \approx 5$   
Assuming top 3 levels are cached:  $5 - 3 = 2$  page reads  
Additional leaf-page scans: 1-2 page reads  
Total Cost:  $\approx 3-4$  page reads

Why Timestamp-only Queries are Avoided

Index order: (user\_id, timestamp)  
Without specifying **user\_id**, MongoDB cannot directly locate matching timestamps.  
Therefore:

- Efficient prefix search is not possible.
- Large portions of the index must be scanned.
- Query optimizer may choose a collection scan instead.

**Conclusion:**

The compound B+ Tree index (**user\_id, timestamp**) efficiently supports queries that include the prefix field **user\_id**. Queries using both **user\_id** and **timestamp range** require only a few page reads and provide excellent performance.

However, queries that use only **timestamp** cannot efficiently utilize the index because the leading field is missing. Therefore, MongoDB may avoid the index and perform a scan instead.

**GITHUB LINK : <https://github.com/sameerreddyvemireddy/DSA-Case-STUDIES>**

| Student Signature | Faculty Signature |
|-------------------|-------------------|
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